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|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| 5(a)(i)  | microtubules ;                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 1 |
| 5(a)(ii) | <p>any <b>two</b> from:</p> <p>ref. to (daughter) chromosomes at poles ; <b>R</b> move to poles <b>I</b> ends</p> <p><b>A</b> (sister) chromatids at poles</p> <p><b>A</b> two separate groups of (daughter) chromosomes</p> <p><b>A</b> two nuclei form (enclosing chromosomes) <b>I</b> there are two nuclei <b>R</b> if in context of cell dividing</p> <p><i>in context of at each pole, so <b>R</b> if context implies only a single nucleus</i></p> <p>nuclear envelope, reassembles / (re-)forms / AW (around chromosomes) ;</p> <p><b>A</b> nuclear membranes reassemble</p> <p>nucleolus / nucleoli, reappear(s) / re-form(s) ;</p> <p>(daughter) chromosomes, become diffuse / become long and thin / decondense / uncoil ;</p> <p><b>A</b> become chromatin</p> <p><b>I</b> chromosomes disappear</p> <p>spindle, disassembles / AW ; <b>A</b> spindle breaks down</p> <p><i>accept ref. to microtubules in a correct context</i></p> | 2 |

| Question | Answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Marks |
|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 5(b)     | <p>any <b>three</b> from:</p> <p>two structural features named ;</p> <p>valves <i>do not count as a feature if described as semilunar</i></p> <p>tunica, externa / adventitia <b>A</b> outer layer with collagen</p> <p>tunica media <b>A</b> middle layer with smooth muscle and elastic tissue</p> <p>thin wall</p> <p>large lumen (relative to wall thickness)</p> <p><i>feature matched to function</i></p> <p><i>valve I semilunar</i></p> <p>to, prevent backflow / help blood move back towards the heart / help move blood in one direction ;</p> <p><i>tunica, externa / adventitia</i></p> <p>contains collagen / thick(est) layer / outer layer, to protect / prevents collapse (from external forces) / helps to maintain shape ; AW</p> <p><i>tunica media</i></p> <p>smooth muscle, for (mechanical) support / to change diameter of lumen / to help move (low / zero, pressure) blood back towards heart ;</p> <p>elastic fibres to accommodate changes in, blood volume / lumen diameter ;</p> <p><b>A</b> allows, vein / blood vessel, to, stretch / recoil</p> <p><b>R</b> stretch and recoil to prevent bursting</p> <p><i>thin walls with large lumen</i></p> <p>holds large(r), volume / quantity / amount, of (low / zero, pressure) blood <b>I</b> more blood</p> <p><b>or</b></p> <p>(large lumen) <i>ref. to</i> minimises / reduces, resistance (from walls) ;</p> | 3     |